NITESH KUMAR MISHRA

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SUMMARY

Generative AI and Backend Developer experienced in building multi-agent, RAG, and LLM-powered applications using Python and FastAPI. Skilled in designing backend systems, REST APIs, databases, and production AI workflows.

EXPERIENCE

AI/Backend Engineering Intern – Techvruk
Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, AI/Computer Vision

- Designed and documented a **microservices architecture** for an AI-powered circuit diagram analysis platform.
- Performed **manual annotation of P&ID datasets** for YOLO-based symbol detection and model training.
- Developed and integrated **FastAPI REST APIs** for diagram upload, processing, authentication, and validation.
- Designed **PostgreSQL/SQLAlchemy database models** and Alembic migrations for companies, users, roles, departments, and diagrams.

PROJECTS

TripMate AI – Multi-Agent Travel Planner Live | GitHub

Tech Stack: FastAPI, LangGraph, Groq, PostgreSQL, MCP

- Built a multi-agent travel planner with a **supervisor-router architecture** dispatching to specialized flight, hotel, weather, and budget agents.
- Integrated **MCP servers** (AviationStack, Tavily, OpenWeather) for live data retrieval and a **Human-in-the-Loop** approval gate via LangGraph interrupts.
- Implemented input guardrails and token-budget-aware LLM calls with retry handling under API rate limits.

CivicAI – AI-Powered Municipal Grievance Platform Live | GitHub

Tech Stack: FastAPI, React, Computer Vision, Multimodal AI

- Designed a **hybrid CV + multimodal AI pipeline** for civic issue detection, severity, and safety-risk assessment.
- Built deterministic backend logic for authority routing, **SLA monitoring**, and automated email escalation.
- Implemented complaint lifecycle tracking, status history, and an authority dashboard with resolution evidence.

Cashora – Personal Finance Platform Live | GitHub

Tech Stack: Python, FastAPI, React, PostgreSQL, LangGraph, Groq

- Built a financial intelligence platform using **React and FastAPI** for automated bank statement analysis and insights.
- Developed an **agentic AI assistant** using LangGraph, RAG, and tool calling for natural-language financial queries.
- Implemented OCR, transaction categorization, financial analytics, and streaming AI responses.

ResonoAI – AI Voice Platform Live | GitHub

Tech Stack: Next.js, TypeScript, PostgreSQL, Docker, AWS

- Engineered an AI-powered SaaS platform with secure authentication and scalable backend architecture.
- Deployed using **Docker, GitHub Actions, AWS EC2, and Nginx** for production workflows.
- Built scalable APIs and PostgreSQL database workflows for reliable handling of user operations.

SKILLS

Languages: Python, C++, JavaScript, TypeScript
AI/GenAI: LLMs, RAG, LangChain, LangGraph, AI Agents, MCP, Vector Databases, Computer Vision
Backend: FastAPI, Node.js, REST APIs, SQLAlchemy, Alembic
Frontend: Next.js, React.js, Tailwind CSS
Databases: PostgreSQL, MongoDB
Cloud & DevOps: Docker, AWS EC2, Nginx, GitHub Actions, Linux

EDUCATION

Maharaja Surajmal Institute of Technology New Delhi, India
B.Tech. – Information Technology

Rajkiya Pratibha Vikas Vidyalaya New Delhi, India
Class 10, Class 12